Throwing Dice

Time limit: 1.00 s
Memory limit: 512 MB

CSES 1096 | Mathematics | https://cses.fi/problemset/task/1096

Your task is to calculate the number of ways to get a sum n by throwing dice. Each throw yields an integer between 1 … 6.

For example, if n=10, some possible ways are 3+3+4, 1+4+1+4 and 1+1+6+1+1.

Input

The only input line contains an integer n.

Output

Print the number of ways modulo 10^9+7.

Constraints

- $1 \le n \le 10^{18}$

Example

Input
8

Output
125